Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2007-I-Q03] | Part A (Q1-Q5)

Answer: 015

1. Expansion gives imaginary parts $18b$ and $243b - b^3$. Equating and dividing by positive b gives $b^2 = 225$.

2. Thus $b = 15$.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2009-I-Q02] | Part A (Q1-Q5)

Answer: 697

1. Write $z = u + 164i$ and clear the denominator: $u + 164i = -656 + 4(u+n)i$.

2. Compare components: $u = -656, u + n = 41$. Thus $n = 697$.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2018-II-Q05] | Part A (Q1-Q5)

Answer: 074

1. Multiplying all equations gives $(xyz)^2 = 240^2(16+30i) = 240^2(5+3i)^2$. Thus $xyz = \pm 240(5+3i)$.

2. Dividing by the three given products gives matching-sign triples $x = \pm(20+12i), y = \pm(-10-10i), z = \pm(-3+3i)$. Both signs satisfy the products.

3. The sum is $\pm(7+5i)$, whose squared modulus is 74.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2022-I-Q04] | Part A (Q1-Q5)

Answer: 834

1. The arguments of w, z, i are $30^\circ, 120^\circ, 90^\circ$. Equality is equivalent to $r + 3 \equiv 4s \pmod{12}$.

2. For $s \equiv 1, 2, 0 \pmod{3}$, the required r residues modulo 12 are $1, 5, 9$. These occur $9, 8, 8$ times in $1, \ldots, 100$; the three s classes have sizes $34, 33, 33$.

3. Therefore the count is $34 \cdot 9 + 33 \cdot 8 + 33 \cdot 8 = 834$.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2026-I-Q05] | Part A (Q1-Q5)

Answer: 065

1. In complex coordinates put $B = 0, A = 1$. Then $A\prime = e^{i\theta}$ and $B\prime = A\prime + e^{-i\theta}(B - A\prime) = e^{i\theta} - 1$.

2. Therefore $AB\prime^2 = |e^{i\theta} - 2|^2 = 5 - 4\cos\theta = 16/9$. Hence $\cos\theta = 29/36$, so $m + n = 65$.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2000-II-Q09] | Part B (Q6-Q10)

Answer: 000

1. The quadratic equation for z has roots $e^{\pm 3^\circ i}$. Therefore $z^{2000} + z^{-2000} = 2\cos 6000^\circ = 2\cos 240^\circ = -1$.

2. The least integer strictly exceeding -1 is 0.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2005-I-Q06] | Part B (Q6-Q10)

Answer: 045

1. Adding one yields $(x-1)^4 = 2006$. The nonreal roots are $1 \pm i\sqrt[4]{2006}$.

2. Their product is $1 + \sqrt{2006}$. Since $44^2 < 2006 < 45^2$, this lies between 45 and 46, so the floor is 45.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2005-II-Q09] | Part B (Q6-Q10)

Answer: 250

1. Write $\sin t + i\cos t = ie^{-it}$. The identity becomes $i^n e^{-int} = ie^{-int}$, equivalent to $n \equiv 1 \pmod 4$.

2. The integers $1, 5, \ldots, 997$ number 250.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2008-II-Q09] | Part B (Q6-Q10)

Answer: 019

1. With $\omega = e^{i\pi/4}$, the move is $z \mapsto \omega z + 10$. Since $\omega^8 = 1$ and the eight powers sum to zero, the affine map also has period eight, so only six moves remain.

2. Thus $z_{150} = 5\omega^6 + 10\sum_{j=0}^{5}\omega^j = -5\sqrt{2} + (5 + 5\sqrt{2})i$. The absolute-coordinate sum is $5 + 10\sqrt{2}$, between nineteen and twenty.

3. The floor is 19.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2010-II-Q07] | Part B (Q6-Q10)

Answer: 136

1. Write $w = x + yi$. The root sum is real, forcing $4y + 12 = 0$. The roots become $x, x + 6i, 2x - 4 - 6i$.

2. The two nonreal roots are conjugate, so $x = 2x - 4 = 4$. Hence $P(1) = (-3)[(-3)^2 + 6^2] = -135$.

3. As $P(1) = 1 + a + b + c$, the required absolute value is 136.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2011-II-Q08] | Part B (Q6-Q10)

Answer: 784

1. The roots have radius eight and arguments $j\pi/6$. Each independent choice contributes $8\max(\cos(j\pi/6), -\sin(j\pi/6))$ to the best real part.

2. For $j = 5, \ldots, 10$, choose the negative sine; for $j = 1, 2, 3, 4, 11, 12$, choose the cosine. Each six-term sum of the normalized selected values is $1 + \sqrt{3}$.

3. The maximum is $16 + 16\sqrt{3} = 16 + \sqrt{768}$, so $m + n = 784$.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2012-I-Q06] | Part B (Q6-Q10)

Answer: 071

1. The imaginary part is positive, so $z \neq 0$. Substitution gives $z^{142} = 1$, whose upper-half-plane roots have arguments $k\pi/71$, $1 \leq k \leq 70$.

2. Sine equality between angles strictly between zero and π means $m/n = k/71$ or $1 - k/71$. Since 71 is prime and k is neither zero nor seventy-one, both fractions reduce with denominator 71.

3. Thus $n = 71$; every such upper root also satisfies the original equations with $w = z^{13}$.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2012-II-Q06] | Part B (Q6-Q10)

Answer: 125

1. The distance factors as $125|1+2i-z^2|$, with $|z^2|=25$. The triangle inequality gives maximum $125(25+\sqrt{5})$ when z^2 points opposite to $1+2i$.

2. Thus $z^2=-25(1+2i)/\sqrt{5}=-5\sqrt{5}(1+2i)$. This nonreal point has a square root in the upper half-plane, so the bound is attained under $b>0$.

3. Squaring yields $z^4=125(-3+4i)=-375+500i$. Hence $c+d=125$.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2012-II-Q08] | Part B (Q6-Q10)

Answer: 040

1. Multiply the two equations and put $v = zw$. The result is $v - 240/v = -30 + 14i$, hence $v^2 + (30 - 14i)v - 240 = 0$.

2. The quadratic formula uses $\sqrt{416 - 210i} = \pm(21 - 5i)$, giving $v = 6 + 2i$ or $-36 + 12i$. Their squared moduli are $40, 1440$.

3. The lower one is attained by $z = 1 - i, w = 2 + 4i$, which satisfy both original equations. Thus the minimum is 40.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2013-I-Q10] | Part B (Q6-Q10)

Answer: 080

1. The conjugate is also a root, while the remaining root $q = a - 2r$ is an integer. Their product gives $q(r^2 + s^2) = 65$, so q is positive and $r^2 + s^2$ is $5, 13$, or 65.

2. The unordered positive square representations are $5 = 1^2 + 2^2$, $13 = 2^2 + 3^2$, $65 = 1^2 + 8^2 = 4^2 + 7^2$. For each, r may be either sign of either component, yielding four distinct polynomials; changing only s sign does not change the polynomial.

3. The root sum is $q + 2r$. The four r values cancel, contributing $4q$ for each representation. The coefficients produced are all nonzero.

4. The total is $4(13 + 5 + 1 + 1) = 80$.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2014-I-Q07] | Part B (Q6-Q10)

Answer: 100

1. The quotient w/z ranges over the radius-$1/10$ circle at the origin, so $(w-z)/z$ ranges over the circle centered at -1 with that radius.

2. Extreme slopes from the origin are tangent to this circle. The right triangle formed by origin, circle center, and tangency point gives $\sin^2\alpha = 1/100$ for the extreme deviation from the real axis.

3. Hence $\tan^2\theta \le (1/100)/(99/100) = 1/99$, with tangency attaining equality. The sum is 100.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2014-II-Q10] | Part B (Q6-Q10)

Answer: 147

1. The nonzero denominators permit multiplication, yielding $z^2 + zw + w^2 = 0$. Hence w/z is one of the two primitive cube roots of unity; both make all original denominators nonzero.

2. The vertices form an equilateral triangle of circumradius 2014. Its area is $(3\sqrt{3}/4)2014^2 = 3 \cdot 1007^2\sqrt{3}$.

3. The coefficient modulo one thousand is $3 \cdot 7^2 = 147$.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2016-I-Q07] | Part B (Q6-Q10)

Answer: 103

1. When $ab + 2016 \geq 0$, the first radical is real. Vanishing imaginary part requires $b = -a$. Then $a^2 \leq 2016$ gives $-44 \leq a \leq 44$, with $a = \pm 10$ excluded by the denominator. This contributes 87.

2. When $ab + 2016 < 0$, the numerator is purely imaginary and must vanish: $-ab - 2016 = |a + b|$. For positive $a + b$, this becomes $(a + 1)(b + 1) = -2015$; for negative sum it becomes $(a - 1)(b - 1) = -2015$. A zero sum cannot satisfy it.

3. The positive divisors of $2015 = 5 \cdot 13 \cdot 31$ are $1, 5, 13, 31, 65, 155, 403, 2015$. In the positive-sum case the positive factor must be one of the last four; either coordinate may carry it, giving 8 pairs. Negating both variables gives another 8 negative-sum pairs. Their denominators are nonzero.

4. Total $87 + 8 + 8 = 103$.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2017-I-Q10] | Part B (Q6-Q10)

Answer: 056

1. The real cross-ratio condition equates directed inscribed angles modulo180 degrees. Since the three fixed points are noncollinear, the locus is their circumcircle, excluding the undefined point z_3.

2. The first two points have perpendicular bisector $y = 61$. The second and third have midpoint $(48, 69)$ and slope1, so their bisector is $y - 69 = -(x - 48)$. The center is therefore $(56, 61)$.

3. The highest point is directly above the center, with real coordinate56. Its height exceeds99, so it is not the excluded point z_3.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2018-I-Q06] | Part B (Q6-Q10)

Answer: 440

1. Write $z = e^{i\theta}$ with $0 \le \theta < 2\pi$. The condition is $\sin 720\theta = \sin 120\theta$, so either $600\theta \equiv 0 \pmod{2\pi}$ or $840\theta \equiv \pi \pmod{2\pi}$.

2. The respective families contain $600, 840$ angles. An overlap would require $\pi k/300 = (2j+1)\pi/840$, or $14k = 5(2j+1)$, contradicting parity.

3. The count is 1440, leaving 440.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2019-II-Q08] | Part B (Q6-Q10)

Answer: 053

1. The input is a sixth root of unity ω, so the powers reduce to $\omega^2, \omega, 1$. Therefore the real and imaginary parts are $c + (b - a)/2$ and $(a + b)\sqrt{3}/2$.

2. Imaginary parts give $a + b = 4038$. Since both coefficients are at most 2019, equality forces $a = b = 2019$. Then $c = 2015$, so $f(1) = 6053$, remainder 53.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2020-I-Q08] | Part B (Q6-Q10)

Answer: 103

1. Identify east with the positive real direction. The displacement multiplier is $z = \frac{1}{2}e^{i\pi/3} = (1 + i\sqrt{3})/4$, whose modulus is less than one.

2. The limiting displacement is $5/(1 - z)$. Since $|1 - z|^2 = (3/4)^2 + (\sqrt{3}/4)^2 = 3/4$, its squared modulus is $25/(3/4) = 100/3$.

3. The required integer is $100 + 3 = 103$.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2023-II-Q08] | Part B (Q6-Q10)

Answer: 024

1. Put $P(t) = t^3 + t + 1$ and let its roots be $\beta_1, \beta_2, \beta_3$. Interchanging the factors in the product gives $\prod_{k=0}^{6} P(\omega^k) = \prod_{j=1}^{3}(1 - \beta_j^7)$, because $t^7 - 1$ has the seven roots ω^k.

2. Reducing powers using $\beta^3 = -\beta - 1$ gives $\beta^7 = 2\beta^2 - 1$. Thus $1 - \beta^7 = 2(1 - \beta)(1 + \beta)$.

3. The product becomes $8P(1)[-P(-1)] = 8 \cdot 3 \cdot 1 = 24$. The negative sign in the second bracket comes from the odd degree of P.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2024-I-Q07] | Part B (Q6-Q10)

Answer: 540

1. Write $z = 4(\cos\theta + i\sin\theta)$. Since $1/z = (\cos\theta - i\sin\theta)/4$, the real part simplifies to $324\cos\theta - 432\sin\theta$.

2. Cauchy–Schwarz bounds it by $\sqrt{324^2 + 432^2} = 540$. Equality is attained at $(\cos\theta, \sin\theta) = (3/5, -4/5)$.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2001-II-Q14] | Part C (Q11-Q15)

Answer: 840

1. The equation is $z^8(z^{20}-1)=1$. On the unit circle this forces $|z^{20}-1|=1$, so $z^{20}=\operatorname{cis}(\pm 60^\circ)$.

2. For the positive sign, $z^8=\operatorname{cis}(-120^\circ)$ and $z^4=z^{20}/(z^8)^2=\operatorname{cis}300^\circ$, giving $\theta\equiv 75^\circ \pmod{90^\circ}$. The negative sign gives $\theta\equiv 15^\circ \pmod{90^\circ}$. Substitution confirms both classes suffice.

3. The ordered arguments are $15, 75, 105, 165, 195, 255, 285, 345$ degrees. The requested sum is $75+165+255+345=840$.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2002-I-Q12] | Part C (Q11-Q15)

Answer: 275

1. Direct composition gives $F^2(z) = i(z+1)/(z-1)$ and $F^3(z) = z$ wherever the orbit is defined. The initial orbit avoids the poles, so it repeats every three steps.

2. As $2002 \equiv 1 \pmod 3$, compute $F(1/137 + i) = (1/137 + 2i)/(1/137) = 1 + 274i$. The sum is 275.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2003-II-Q15] | Part C (Q11-Q15)

Answer: 015

1. The triangular coefficient pattern is the convolution of two length-24 all-one lists, so $P(x) = x(1 + x + \cdots + x^{23})^2$.

2. Its distinct zeros are 0 and $e^{15k^\circ i}$ for $k = 1, \ldots, 23$. Thus the required terms are $|\sin 30k^\circ|$; the zero root contributes nothing.

3. Each twelve-step absolute-sine cycle sums to $4 + 2\sqrt{3}$. Over the twenty-three terms the absent twenty-fourth term is zero, so the total is $8 + 4\sqrt{3}$, giving 15.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2004-I-Q13] | Part C (Q11-Q15)

Answer: 482

1. Multiply by $(x-1)^2$: the result is $(x^{18}-1)^2 - x^{17}(x-1)^2 = (x^{19}-1)(x^{17}-1)$. The two roots at one cancel, while $P(1) = 323 \neq 0$.

2. Thus the roots are all nontrivial seventeenth and nineteenth roots of unity. The first five arguments are $1/19, 1/17, 2/19, 2/17, 3/19$.

3. Their sum is $6/19 + 3/17 = 159/323$, giving 482.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2008-II-Q13] | Part C (Q11-Q15)

Answer: 029

1. The exterior is the union of six rotated half-planes such as $\Re z > 1/2$. Under $w = 1/z$, that inequality becomes $u/(u^2 + v^2) > 1/2$, or $|w - 1| < 1$. Thus S is the union of six unit disks centered at the sixth roots of unity.

2. Partition into sixty-degree wedges centered on the disk-center rays. In each wedge its corresponding disk reaches farthest in every direction, with boundary endpoints $(3/2, \pm\sqrt{3}/2)$ for the wedge around the real axis.

3. That wedge piece has a 120° unit sector of area $\pi/3$ plus two triangles of total area $\sqrt{3}/2$. Multiply by six: area $2\pi + 3\sqrt{3} = 2\pi + \sqrt{27}$, giving 29.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2009-II-Q13] | Part C (Q11-Q15)

Answer: 672

1. Use complex coordinates $A=-2, B=2, C_i=2\omega^i$, with $\omega=e^{i\pi/7}$. Then $AC_i\cdot BC_i=4|1-\omega^{2i}|$.

2. The numbers ω^{2i} are the six nontrivial seventh roots. Evaluating $(x^7-1)/(x-1)$ at one gives $\prod_{i=1}^{6}|1-\omega^{2i}|=7$.

3. Thus $N=4^6\cdot 7=28672$, remainder 672.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2012-I-Q14] | Part C (Q11-Q15)

Answer: 375

1. Vieta gives $a+b+c=0$. Suppose the right angle is at b, with hypotenuse length $h=|a-c|$. Its midpoint $(a+c)/2=-b/2$ is the circumcenter, so $3|b|/2=h/2$, or $h=3|b|$.

2. The parallelogram identity gives $|a|^2+|c|^2=[|a-c|^2+|a+c|^2]/2=(h^2+|b|^2)/2$.

3. Hence $250=h^2/2+3|b|^2/2=(2/3)h^2$, yielding $h^2=375$.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2013-I-Q14] | Part C (Q11-Q15)

Answer: 036

1. The absolutely convergent complex series is $Q + iP = \sum_{k \geq 0} (ie^{i\theta}/2)^k = 2/(2 - ie^{i\theta})$.

2. Rationalizing gives $P/Q = \cos\theta/(2 + \sin\theta)$; the denominator is positive. Squaring the given equality yields $49(1 - s^2) = 8(2 + s)^2$ for $s = \sin\theta$.

3. This factors as $(3s - 1)(19s + 17) = 0$. The interval requires $s \leq 0$, so $s = -17/19$. The original positive ratio then chooses $\cos\theta = 6\sqrt{2}/19$, a valid fourth-quadrant value.

4. The answer is $17 + 19 = 36$.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2017-II-Q12] | Part C (Q11-Q15)

Answer: 110

1. Use complex coordinates with initial center0 and $A_0 = 1$. Let O_n denote the centers. Induction gives $A_n - O_n = r^n i^n$; internal tangency then gives $O_{n+1} - O_n = (1-r)(ir)^n$.

2. The radii shrink to zero in nested disks, so their unique common point is the center limit $B = (1-r)/(1-ir)$. Its distance from0 is $(1-r)/\sqrt{1+r^2}$.

3. At $r = 11/60$, this equals $49/\sqrt{3600+121} = 49/61$. Thus $m+n = 110$. The center-to-limit distance after stage n is $r^n(1-r)/\sqrt{1+r^2} < r^n$, so the point is inside every disk.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2019-I-Q12] | Part C (Q11-Q15)

Answer: 230

1. The consecutive displacement ratio is $[f(f(z)) - f(z)]/[f(z) - z] = (z - 19)(z + 1)$, using $f(z) - z = z(z - 20)$. Perpendicular nonzero legs require this ratio purely imaginary and nonzero.

2. For $z = x + 11i$, its real part is $x^2 - 18x - 140$. Set zero to obtain $x = 9 \pm \sqrt{221}$. The positive specified form selects $x = 9 + \sqrt{221}$, and its ratio has nonzero imaginary part.

3. Thus $m + n = 9 + 221 = 230$.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2023-I-Q15] | Part C (Q11-Q15)

Answer: 349

1. Except for $p = 2$, write $p = a^2 + b^2$ with integers $a > b > 0$. Multiplication by units and conjugation allow signs and order of the real and imaginary cube components to be adjusted; their positive magnitudes are therefore the candidate sides.

2. For $X = a^3 - 3ab^2, Y = 3a^2b - b^3$, factor $X + Y = (a - b)(p + 4ab)$ and $X - Y = (a + b)(p - 4ab)$. Since the first magnitude exceeds p, the triangle condition is exactly $(a + b)|p - 4ab| < p$. In particular $|a^2 + b^2 - 4ab| < \sqrt{p}$.

3. The finite bounds are $1 \le b \le 21$ and $b < a \le 31$. Testing a for each such b in $(a^2 + b^2 - 4ab)^2 < a^2 + b^2 < 1000$ gives possible values
$10, 17, 53, 68, 130, 153, 212, 241, 272, 349, 386, 520, 565, 725, 778, 905, 964$. The primes among them are $17, 53, 241, 349$; their $(a + b)|p - 4ab|$ values are $5, 27, 19, 253$, all smaller than p.

4. The largest is 349. Indeed $z = 18 + 5i$ has $z^3 = 4482 + 4735i$, and $4735 - 4482 = 253 < 349 < 4482 + 4735$.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2024-II-Q13] | Part C (Q11-Q15)

Answer: 321

1. Factor $t^2 - 2t + 2 = (t - (1 + i))(t - (1 - i))$. Since 13 is prime, the powers of ω exhaust the roots of $t^{13} - 1$. The product is therefore $[1 - (1 + i)^{13}][1 - (1 - i)^{13}]$.

2. As $(1 + i)^{13} = -64 - 64i$, this becomes $(65 + 64i)(65 - 64i) = 65^2 + 64^2 = 8321$. The remainder is 321.

Source / 来源

Section 1.3 Complex numbers & roots of unity

复数与单位根

[AIME-2026-II-Q11] | Part C (Q11-Q15)

Answer: 132

1. Choose nonzero square roots s_1, s_2, s_3 of the roots. The eight sign sums come in opposite pairs. Exactly seven values therefore requires a zero sign sum; coincidences between nonopposite choices force $s_i = \pm s_j$ and would reduce the count to at most six.

2. The product of the four essentially different sign sums is $4e_2 - e_1^2$, where e_1, e_2 are elementary symmetric functions of s_i^2. Its vanishing gives $4(n-11) - (n/6)^2 = 0$, so $n = 12$ or 132.

3. At $n = 132$, factor the cubic as $(x-16)(x^2 - 6x + 25)$ and choose roots $4, 2+i, 2-i$. The sign sums are zero twice and six distinct values $\pm 8, \pm(4+2i), \pm(4-2i)$. Thus 132 qualifies and is greatest.

Source / 来源